DIIC Assignment 2 - Group 6

LAB 1 - Tuesday 9:30h - 12:30h

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Design Context: Deaf adults - Hospital

HMW Statement

How might we support deaf adult patients to communicate effectively and independently with medical practitioners in a hospital, to ensure accurate diagnoses, maintain medical confidentiality, and a safe healthcare environment?

Ideas (Brainstorm)

1. Two-way translation glasses for the deaf. Transforming audio into sign language/text and vice-versa.
2. Translation tablet for every deaf patient that translates speech and sign language in real time.
3. Sign to speech gloves, transforms sign language into audible language.
4. Digital body maps: Allows deaf patients to use a body diagram to show pain and symptoms effectively to the doctor.
5. A panel (next to the patient's bed) that flags which patients are deaf or hard of hearing and their preferred communication method.
6. Interactive timeline allowing deaf adult patients to temporally identify (how long it's been happening, when it started, etc.) their symptoms.
7. Video remote interpreter (more reliable than using AI) but has to be available.
8. Accessible check-in application/machine allows deaf adults to independently make appointments and complete their registration.
9. Instead of sound (alarms etc.) to signalize emergencies, light switch patterns.
10. Real-Time AI Sign Language Translator: A consultation camera-based AI system that translates the patient's sign language into real-time text or voice for the doctor.
11. 3D Sign Language Avatar Display: Converts the doctor's spoken voice in real time into a 3D avatar that translates the dialogue into sign language on the patient's screen.
12. Visual Medical Explainer Software: A program that converts complex medical explanations into dynamic visual diagrams and interactive infographics.
13. Interactive Anatomical Pain Mapping Interface: An interface featuring a 3D anatomical model that allows patients to specify the location, duration, type, and intensity of their pain.
14. On-demand sign language interpreter. A simple call/button connects to a remote (or physical) sign language interpreter immediately.
15. Hybrid interpretation system: AI translates most information, maintaining confidentiality but a human translator effectively describes the medical terms to the adult deaf patient (only when the AI system fails to do so effectively).
16. Communication Dashboard: Allows medical practitioners to know the preferred communication method before the appointment even starts.
17. Visual medication guides: Medical practitioners can describe medication usage using guides and infographics, displaying timing, dosage and warnings.

18. When standing in line, instead of calling the name of the patient, flag their number on a screen or have a timer.
19. Procedure preview - Deaf adult patients can watch a quick visual explanation of what procedure is about to be done to them, as well as what parts of their body may be examined.
20. Global healthcare communication card - Allows deaf adult patients to maintain a universal communication card/profile, usable at all hospitals, indicating their needs and preferences.
21. Vibration based queue turn - A small bracelet given to deaf adults when they do the check-in, allows them to know when the medical practitioner is ready by vibrating and showing a visual cue.
22. Vibrating bracelet that vibrates in case of medical emergencies.
23. Communication check - Deaf adult patients are given a button/interactive way to display their consent or to confirm they understood an important message given by the medical practitioner.
24. App that helps medical practitioners learn/practice how to interact with hard of hearing patients.
25. Sign Language Pre-Consultation Kiosk: A private booth in the waiting area where the patient records the reason for their visit in Sign Language; AI then generates a concise clinical text summary sent directly to the doctor's computer.
26. Digital NFC Prescription Card: A digital card handed to the patient during the consultation that, when tapped against a smartphone, plays a Sign Language video explaining the dosage, schedule, and potential side effects of the prescribed medications.
27. QR code on the patient's bracelet that gives all information of the patient to the practitioner.
28. Neuro-Interactive Interface (BCI Light): A headband that detects the patient's communication intentions during moments of high stress or intense pain.
29. Tool that helps medical practitioners learn how to effectively treat and communicate with adult deaf patients, ensuring they know basic sign language, correct behaviors like speaking directly to the deaf patient instead of the companion, and patience, allowing both the patient and the practitioner to have a private, helpful and successful medical appointment.
30. Medical practitioners can use a special mask that mirrors their lips, ensuring lip-reading capabilities are maintained in these environments.
31. App where the patient is able to preemptively "book" an interpreter to be with him during the consultation.
32. Two pairs of glasses, one for the patient and other for the doctor. The patient ones have a microphone to listen to the doctor and display the text inside the glasses. The doctors see the sign language and display text or output the audio.
33. A mini projector that the doctor has on his table. The projector has a camera pointing to the patient and an audio input near the doctor, then processes the image and audio and projects the conversation onto the table or a nearby wall.
34. A bracelet that can keep track of the hand movement of the patient doing sign language, and has an audio output to communicate with the practitioner.
35. An app with the hospital environment integrated that allows the patient to be guided and interact, minimizing the actual person-to-person contact needed.

36. A robotized medical assistant that can play the role of an intermediary between the patient and doctor without the need of a relative or assistant worker, maintaining the privacy and independence of the patient.
37. An app to be used on a tablet to communicate with the patient and receive easy feedback, especially for patients in worse conditions. The app allows the doctor to make questions and, with the help of AI, generate possible answers so that the patient just has to click on what they want.
38. For a medical intervention, a fixed display where the instructions given by the doctor appear written.
39. An interactive board where both the doctor and patient can write and the conversation can flow with the help of graphics and images.

Selection Process

Step 1: Thematic Clustering

All ideas were grouped into four functional touchpoints: Check-in/Waiting Area, Patient Expression, Doctor Explanation, and Post-Consultation Follow-up.

Pre-Consultation, Check-In & Arrival

- Sign Language Pre-Consultation Kiosk
- Accessible Check-In Kiosk/App
- Vibration-Based Queue Bracelet
- Visual Queue Display System
- Preemptive Interpreter Booking App
- Integrated Hospital Navigation App
- Global Healthcare Communication Card

In-Consultation: Patient Expression (Patient -> Doctor)

- Real-Time AI Sign Language Translator
- Interactive Anatomical Pain Mapping Interface
- Interactive Symptom Timeline
- Hand-Movement Tracking Bracelet
- Sign-to-Speech Gloves
- Neuro-Interactive Interface (BCI Light)

In-Consultation: Doctor Explanation (Doctor -> Patient)

- 3D Sign Language Avatar Display
- Visual Medical Explainer Software
- Procedure Preview Tool
- Smart Masks
- Communication Check Button

Dedicated Translation Devices & Hardware

- Two-Way Translation AR Glasses
- Dual-Pair Doctor-Patient Glasses
- Bedside/Desk Translation Tablet

- Tabletop Mini-Projector System

Human Interpretation & Hybrid Models

- Hybrid AI-Human Interpretation System
- Anonymous Remote Interpreter App
- Video Remote Interpreter (VRI) On-Demand

Post-Consultation, Environment & Practitioner Training

- Digital NFC Prescription Card
- Visual Medication Guides
- Doctor Communication Dashboard
- Patient Bedside Communication Panel
- Emergency Light Switch Patterns
- Emergency Vibrating Wristband
- Patient Wristband QR Code
- Medical Practitioner Communication Training App

Step 2: Prioritization & Elimination

Through analysis and team voting, we evaluated all concepts against four core criteria: **Privacy**, **Autonomy**, **Medical Accuracy**, and **Feasibility**, to select solutions that directly address the core consultation barrier rather than pre- or post-consultation phases: ensuring the patient can unambiguously express their symptoms and fully comprehend the doctor while maintaining absolute confidentiality.

Final Chosen Ideas:

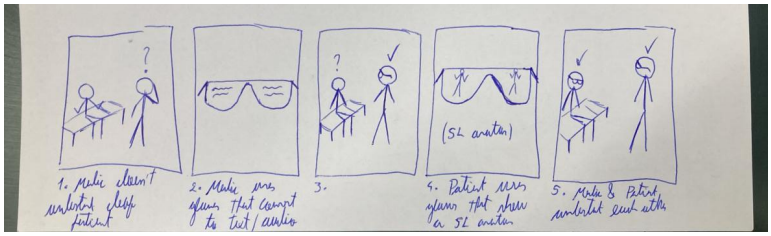
- **(15) Hybrid interpretation system:** AI translates most information, maintaining confidentiality, but a human translator effectively describes the medical terms to the adult deaf patient (only when the AI system fails to do so effectively).
- **(32) Two pairs of glasses**, one for the patient and other for the doctor. The patient glasses have a microphone to listen to the doctor and display the text inside the glasses. The practitioner glasses see the sign language and display the text and output it into audio.
- **(17), (12) Visual guides:** Medical practitioners can describe medication usage using guides and infographics, displaying timing, dosage and warnings, as well as explain complex medical terminology using pictures and diagrams, ensuring that a lack of direct translation still results in successful communication.

Step 3: Final Solution Synthesis

Rather than choosing a single idea, we combined three into a device that could do all the work required. The base idea is the glasses that allow the doctor and patient to understand each other through real-time interpretation integrated into the glasses. To achieve this we merged the idea of an AI translator to sign language, so that deaf people who are not fluent in reading have a better option to understand what the doctor is saying. Additionally, through the glasses added software, graphics and images can be shown to better explain specific medical concepts. In the doctor's glasses, the AI translator is also used to allow the sign language to be converted into audio that the doctor can listen to through the glasses speaker.

Storyboards

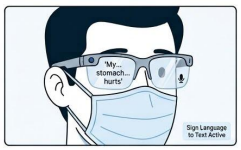
Sketch:



AI Improved:



1. Arrival & Misunderstanding (Expressing)
A deaf patient arrives and uses Sign Language to describe symptoms, but the doctor cannot understand, establishing an immediate communication barrier.



2. Doctor's Solution
The doctor puts on special smart glasses that recognize and translate Sign Language into real-time text overlays, enabling partial expression and diagnosis.



3. Perception Barrier (Perceiving)
The doctor can now speak naturally, but the patient cannot understand the verbal instructions and diagnosis, creating a complete barrier from the patient's side.



4. Patient's Solution. The patient wears matching smart glasses. A real-time 3D avatar appears in their field of view, translating the doctor's voice into understandable Sign Language.



5. Mutual Understansing. With both wearing the dual-smart-glasses translation ecosystem, the consultation proceeds with seamless, private, independent two-way communication, ensuring accurate diagnosis and full patient autonomy.